Tutorials and examples

SWAT+ Documentation

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Contents

Pages . . . 1
Prerequisites . . . 2
file.cio: the index of every other input . . . 2
simulation group . . . 2
basin group . . . 3
climate group . . . 3
connect, hru, hydrology groups . . . 4
hru_parm_db, ops, lum groups . . . 4
init, soils, decision_table groups . . . 5
carbon database . . . 5
Run the model . . . 5
Read the outputs . . . 5
What to change next . . . 6
Prerequisites . . . 6
file.cio: every group used . . . 7
time.sim and print.prt . . . 7
codes.bsn . . . 7
climate . . . 7
hru.con: a single HRU . . . 8
hru-data.hru . . . 8
landuse.lum and management.sch . . . 8
aquifer.aqu and aquifer.con . . . 8
channel-lte.cha and hyd-sed-lte.cha . . . 9
Run the model . . . 9
Read the outputs . . . 9
What to change next . . . 9
What it reads . . . 10
What it draws . . . 10
How to run it . . . 10
What you should see . . . 11
Related . . . 11
Add a new HRU . . . 11
Swap the PET method . . . 11
Switch to Green-Ampt infiltration . . . 11
Change the simulation period . . . 12
Enable the carbon model . . . 12

This section takes the small datasets that ship in `swatplus/refdata/` and walks them apart piece by piece. Each tutorial assumes you have installed SWAT+ and read the Getting Started pages.

Pages

- Ames_sub1 walkthrough. A one-subbasin, twelve-HRU watershed near Ames, Iowa, simulated 1975 to 2020. Open every input group, run, and read the outputs.

- Osu_1hru walkthrough. A single-HRU paddy near Osu, Korea, simulated 2010 to 2023. Same structure as Ames at the smallest possible scale.
- Carbon notebook. The Jupyter notebook in `swatplus/jupyter_notebooks/carbon.ipynb` that plots the soil-carbon outputs.
- Cookbook recipes. Short, copy-and-paste edits: add an HRU, switch the PET method, switch to Green-Ampt, change the simulation period, turn on the carbon model.

`Ames_sub1` is the smallest realistic SWAT+ project in the source tree. One subbasin near Ames, Iowa. Twelve HRUs. Forty-six years of daily climate (1975 to 2020). The folder is at `swatplus/refdata/Ames_sub1`.

This page opens every file in the project, in the order SWAT+ reads them, explains what is in it, and then runs the model and reads the outputs. Use it as the second pass over a SWAT+ project after the Quick start.

Prerequisites

- A working SWAT+ executable. See Install.
- A local copy of `refdata/Ames_sub1`.

file.cio: the index of every other input

`file.cio` tells SWAT+ which other files exist. It is the first file the model reads. The literal token `null` means "this slot is not used in this project".

Open `Ames_sub1/file.cio`. The relevant rows for this project are:

```
simulation        time.sim          print.prt         null              object.cnt        null
basin             codes.bsn         parameters.bsn
climate           weather-sta.cli   weather-wgn.cli   null              pcp.cli
 ↪  tmp.cli
connect           hru.con
hru               hru-data.hru
hydrology         hydrology.hyd     topography.hyd
hru_parm_db       plants.plt        fertilizer.frt    tillage.til       pesticide.pes     ...
 ↪  snow.sno
ops               harv.ops          graze.ops         irr.ops           chem_app.ops
 ↪  fire.ops   sweep.ops
lum               landuse.lum       management.sch    cntable.lum       cons_practice.lum
 ↪  ovn_table.lum
init              plant.ini         soil_plant.ini
soils             soils.sol         nutrients.sol
decision_table    lum.dtl
regions           ls_unit.ele       ...  aqu_catunit.ele
```

Most slots are `null`. Ames_sub1 has no channels, no reservoirs, no routing units, no aquifer file. It is a pure HRU water-balance run. The `connect` row points to `hru.con` only.

Reference: file.cio.

simulation group

time.sim

```
day_start  yrc_start   day_end   yrc_end      step
        0       1975         0      2020         0
```

`step = 0` means a daily time step. `day_start = 0` and `day_end = 0` mean run from January 1 of `yrc_start` to December 31 of `yrc_end`.

Reference: time.sim.

print.prt

The print configuration. The first block sets the warm-up and date window:

```
nyskip      day_start  yrc_start  day_end   yrc_end   interval
0           0               1975         0      2020          1
```

`nyskip = 0` means no warm-up years are discarded from output. The objects table that follows toggles each output file. In Ames_sub1, every row is set to `aa` only:

```
objects                   daily       monthly        yearly         avann
basin_wb                      n             n             n             y
hru_wb                        n             n             n             y
...
```

That is why the run produces `*_aa.txt` files and no daily or monthly outputs.

Reference: print.prt.

object.cnt

Counts of objects to allocate. SWAT+ reads this once and sizes its arrays accordingly.

Reference: object.cnt.

basin group

codes.bsn

Method switches that apply to the whole basin. Open `codes.bsn` and read row 3:

```
pet  event  crack  swift_out  sed_det  rte_cha  deg_cha  wq_cha  nostress  cn  c_fact  carbon
 ↪  lapse  uhyd  sed_cha  tiledrain  wtable  soil_p  gampt  atmo_dep  stor_max  i_fpwet
 ↪  gw_flow  idc_till
  2      0      0          0        0        0        0       0         0   2       0       1
 ↪  0     1        0          0       0       0      0         0         0        0        0
 ↪  3
```

Notable settings for this project:

- `pet = 2`: Hargreaves PET.
- `carbon = 1`: reserved C-FARM slot, which runs no carbon model. The dynamic carbon model now activates on `carbon = 2`; this dataset must be changed to `carbon = 2` to run it.
- `uhyd = 1`: gamma-function unit hydrograph.
- `gampt = 0`: curve-number infiltration (the default).

Reference: codes.bsn.

parameters.bsn

Numeric constants used basin-wide (sediment routing coefficients, plant-water-stress thresholds, surface-runoff lag, and so on). The defaults in this file are the same defaults a fresh QSWAT+ project produces.

Reference: parameters.bsn.

climate group

The Ames station has only two climate files: `ames.pcp` (precipitation) and `ames.tem` (temperature, legacy `.tem` extension). The `weather-sta.cli` file lists one station, `weather-wgn.cli` holds the monthly statistical generator for it, and `pcp.cli` and `tmp.cli` are index files that point at `ames.pcp` and `ames.tem` respectively.

PET is computed by Hargreaves (`pet = 2` above), so there is no PET file. Solar radiation, humidity, and wind are generated from the weather generator.

References: *.cli, *.pcp, *.tmp, *.tem.

connect, hru, hydrology groups

hru.con

The connectivity file for HRU objects. One row per HRU with its area, coordinates, elevation, and the object the HRU drains to. In Ames_sub1 every HRU drains to itself (`hyd_typ = tot` to a downstream object of `null`), so the basin output is the sum of HRU contributions with no routing.

Reference: *.con.

hru-data.hru

The pointer table. Twelve rows, one per HRU. Each row names the topography record, hydrology record, soil, land-use management, soil/plant initialization, surface storage, snow, and field for that HRU:

```
id  name      topo          hydro     soil      lu_mgt           soil_plant_init  surf_stor
 ↪  snow     field
 1  hru0001   topohru0001   hyd0001   soil_01-h1   cosy_lum      soilplant1       null
 ↪  snow001  null
 2  hru0002   topohru0002   hyd0002   soil_02      mntill_corn_lum soilplant1     null
 ↪  snow001  null
...
12  hru0012   topohru0012   hyd0012   soil_12      cosy_lum      soilplant1       null
 ↪  snow001  null
```

Eleven of the twelve HRUs are `cosy_lum` (corn-soy rotation). HRU 2 is `mntill_corn_lum` (minimum-till continuous corn). That single contrast is the point of the dataset: the other HRUs differ only in soil.

Reference: hru-data.hru.

hydrology.hyd, topography.hyd

Per-HRU hydrology and topography parameters. `hydrology.hyd` carries the curve number adjustment, lateral and tile flow coefficients, perco coefficient, ESCO, EPCO, and so on. `topography.hyd` carries slope, slope length, and field length.

hru_parm_db, ops, lum groups

These are the parameter and operation databases the HRUs and management schedules look things up in.

- `plants.plt`: plant-growth parameters per land cover code.
- `fertilizer.frt`, `tillage.til`, `pesticide.pes`, `urban.urb`, `septic.sep`, `snow.sno`: input databases.
- `harv.ops`, `graze.ops`, `irr.ops`, `chem_app.ops`, `fire.ops`, `sweep.ops`: management operation databases.
- `landuse.lum`: each row binds one `lu_mgt` name (used in `hru-data.hru`) to a CN, a management schedule, a conservation practice, and an n-table.
- `management.sch`: ordered list of operations per management name. The Ames dataset has two: `cosy_lum` (corn-soy rotation) and one continuous-corn variant.
- `cntable.lum`, `cons_practice.lum`, `ovn_table.lum`: lookups referenced by `landuse.lum`.

References: plants.plt, fertilizer.frt, tillage.til, snow.sno, *.ops, landuse.lum.

init, soils, decision_table groups

- `plant.ini` and `soil_plant.ini`: starting conditions for plant biomass, residue, soil water content, soil nutrient pools.
- `soils.sol` and `nutrients.sol`: layered soil profiles and per-layer nutrient pools for each named soil.
- `lum.dtl`: decision tables that govern automatic operations (planting, harvesting) inside the management schedule.

References: soils.sol, *.dtl.

carbon database

The dataset ships `carbon = 1` in `codes.bsn`, which is now the reserved C-FARM slot and runs no carbon model. Set `carbon = 2` to activate the dynamic CENTURY/SWAT-C model; coefficients then come from `carbon.bsn` and `carbon_lyr.bsn`, and initial carbon pools from `soil_plant.ini`.

Reference: carb_coefs.cbn.

Run the model

From inside the `Ames_sub1` folder:

`./swatplus-<version>-linux-x86_64`

The console will fill with `Original Simulation  mm  dd  yyyy Yr N of 46`. The full run takes a few seconds on a recent laptop.

Read the outputs

When the run finishes, the folder contains a new set of files. Start with:

simulation.out

The last lines should be:

```
Original Simulation       12  31  2020 Yr   46 of   46 Time   8:49:53

Execution successfully completed
```

If you do not see `Execution successfully completed`, the run aborted. Look at the last simulated day printed and check `diagnostics.out`.

diagnostics.out

Every line here is a missing reference. The shipped run prints:

```
DIAGNOSTICS.OUT FILE
corn_lum                CORN_rot     not found in management.sch
agrr_lum                AGRR_rot     not found in management.sch
soyb_lum                SOYB_rot     not found in management.sch
oats_lum                OATS_rot     not found in management.sch
wwht_lum                WWHT_rot     not found in management.sch
ames.tem                             file not found (tgage)
                                     file not found (basins_carbon.tes)
```

The first five lines are land-use entries in `landuse.lum` that reference schedules not present in `management.sch`. They are unused in the run (none of the HRUs use those `lu_mgt` names), so the outputs are still correct. The `ames.tem` line refers to a station gage slot in `weather-sta.cli` that was set; the file exists, but the reader did not match it to the `tgage` slot. `basins_carbon.tes` is an optional carbon test file.

Reference: Checker and diagnostics.

checker.out

One row per HRU of the resolved soil parameters:

```
sname     hydgrp  zmx(mm)  usle_k  sumfc(mm)  sumul(mm)  usle_p  usle_ls  esco   epco   cn3_swf
 ↪  perco  latq_co  tiledrain
soil_01   B       2000.0   0.0     361.5      616.18     1.0     0.3638   0.95   1.0    0.0
 ↪  0.5    0.01     0
soil_02   B       2000.0   0.0     361.5      445.67     1.0     0.3638   0.95   1.0    0.0
 ↪  0.5    0.01     0
...
```

In this dataset `usle_k = 0.0` on every HRU. That is a known artifact of the Ames soil definitions; sediment yield will be zero. The `zmx` is 2000 mm on every HRU; the soil rooting depth is taken from `soils.sol`. The `tiledrain` flag is `0` on every HRU.

basin_wb_aa.txt

The basin-average annual water balance. Columns are documented in Water balance outputs. A typical line for Ames_sub1:

```
precip   snofall  snomlt   surq_gen  latq   wateryld  perc   et       pet
1905.2   220.7    219.997  59.5      0.06   59.6      0.003  1846.4   3464.5
```

Precipitation is 1905 mm, ET is 1846 mm, percolation is essentially zero, and surface runoff is 60 mm. The result is dominated by ET because the basin is small and well-drained and the simulation is at HRU scale only.

Other outputs

- `hru_wb_aa.txt`: same fields as `basin_wb_aa.txt`, one row per HRU.
- `hru_nb_aa.txt`, `hru_ls_aa.txt`, `hru_pw_aa.txt`: nutrient balance, landscape balance, plant and water stress, per HRU.
- `crop_yld_aa.txt`: average annual yield by HRU and crop.
- `basin_carbon_aa.txt`, `hru_carbon_aa.txt`: carbon outputs (only when `carbon = 2` activates the carbon model).

What to change next

- Switch the PET method by editing `codes.bsn` (`pet = 0` for Priestley-Taylor, `pet = 1` for Penman-Monteith). See the cookbook.
- Turn on daily output by setting `daily` to `y` in `print.prt` for the rows you care about.
- Change `nyskip` in `print.prt` to discard the first few years as warm-up.
- Read the Carbon notebook for plotting the carbon outputs.

`Osu_1hru` is the smallest SWAT+ project that exercises every input group. One HRU, ten hectares, near Osu, Korea. The land use is `rice140_lum` (a 140-day rice rotation). The folder is at `swatplus/refdata/Osu_1hru`.

Use this dataset when you want to see every input file in `file.cio` populated, but only one HRU to track.

Prerequisites

- A working SWAT+ executable. See Install.
- A local copy of `refdata/Osu_1hru`.

file.cio: every group used

Unlike Ames_sub1, this project populates almost every slot. Open `Osu_1hru/file.cio`:

```
simulation        time.sim          print.prt         null              object.cnt
basin             codes.bsn         parameters.bsn
climate           weather-sta.cli   weather-wgn.cli   null              pcp.cli
 ↪  tmp.cli           slr.cli           null              wnd.cli
connect           hru.con           null              rout_unit.con     null
 ↪  aquifer.con       null              null              null      recall.con
 ↪  null              null              null              chandeg.con
channel           initial.cha       null              null              null
 ↪  nutrients.cha     channel-lte.cha   hyd-sed-lte.cha
reservoir         initial.res       null              null              sediment.res
 ↪  nutrients.res     weir.res          wetland.wet       hydrology.wet
routing_unit      rout_unit.def     rout_unit.ele     rout_unit.rtu
hru               hru-data.hru
aquifer           initial.aqu       aquifer.aqu
hydrology         hydrology.hyd     topography.hyd    field.fld
```

The project has channels, aquifers, routing units, reservoirs, recall, and a wetland. Most of those tables are populated with template entries from the SWAT+ Editor; the single HRU in `hru.con` is what actually runs.

Reference: file.cio.

time.sim and print.prt

`time.sim`:

```
day_start  yrc_start   day_end   yrc_end      step
        1       2010       365      2023         0
```

January 1 2010 through December 31 2023, daily.

`print.prt` has `nyskip = 3`, so the first three years are discarded. The objects table sets `basin_wb` to `yearly` and `avann`. Every other object is `avann` only.

References: time.sim, print.prt.

codes.bsn

```
pet  event  crack  rtu_wq  sed_det  rte_cha  deg_cha  wq_cha  nostress  cn  c_fact  carbon
 ↪  lapse  uhyd  sed_cha  tiledrain  wtable  soil_p  gampt  atmo_dep  stor_max  i_fpwet
  1      0      0       1         0        0        0       1         0   0       0       0
 ↪  0     1        0          0       0       0      0          a        0        0
```

Settings worth noting:

- `pet = 1`: Penman-Monteith PET. Needed because the climate has solar radiation, humidity, and wind.
- `rtu_wq = 1`: routing-unit water quality on.
- `wq_cha = 1`: in-stream water quality on.
- `carbon = 0`: legacy static mineralization (no C-FARM, no Century).
- `gampt = 0`: curve-number infiltration.

Reference: codes.bsn.

climate

Five climate files for the Imsil station:

- `Imsilpcp.pcp`: daily precipitation.
- `Imsiltmp.tmp`: daily max and min temperature.
- `Imsilsol.slr`: daily solar radiation.
- `Imsilwind.wnd`: daily wind speed.

(Relative humidity is generated from the weather generator.) The index files `pcp.cli`, `tmp.cli`, `slr.cli`, `wnd.cli` each list one filename, and `weather-sta.cli` binds those four index files to the station `s35610n127290e`.

References: *.cli, *.pcp, *.tmp.

hru.con: a single HRU

```
id  name     gis_id  area  lat       lon         elev    hru  wst              cst  ovfl  rule
 ↪  out_tot
 1  hru0001       1    10  35.65311  127.36763   350.97   1   s35610n127290e   0    0     0
 ↪  0
```

Area is 10 ha. `hru = 1` and `wst = "s35610n127290e"` bind this row to HRU 1 and to the weather station. `out_tot = 0` means no downstream routing object: the run produces HRU-scale outputs only.

Reference: *.con.

hru-data.hru

The single HRU points at:

```
id  name      topo          hydro      soil        lu_mgt        soil_plant_init  surf_stor
 ↪  snow     field
 1  hru0001   topohru0001   hyd0001    PadHOEGOG   rice140_lum   soilplant1       paddy0001
 ↪  snow001  null
```

- `soil = PadHOEGOG`: a paddy soil. See `soils.sol`.
- `lu_mgt = rice140_lum`: maps in `landuse.lum` to plant community `rice140_comm` and management schedule `rice140_rot`.
- `surf_stor = paddy0001`: surface-storage record for ponded water. This is what makes it a paddy.

Reference: hru-data.hru.

landuse.lum and management.sch

`landuse.lum` row for `rice140_lum` binds the HRU to:

- `plnt_com = rice140_comm` (plant community in `plant.ini`).
- `mgt = rice140_rot` (management schedule in `management.sch`).
- `cn2 = rc_strow_g` (curve-number table entry in `cntable.lum`).
- `cons_prac = cross_slope`.
- `ov_mann = convtill_res`.

`management.sch` then describes the 140-day rice rotation: paddy-fill, transplant, fertilize, harvest, drain.

Reference: landuse.lum.

aquifer.aqu and aquifer.con

`aquifer.aqu` defines twenty shallow aquifer records (`aqu011` through `aqu102`) plus one deep aquifer (`aqu_deep010`). They share identical parameters. The single HRU links to one of them through `aquifer.con` so the percolation has somewhere to go.

Reference: *.aqu.

channel-lte.cha and hyd-sed-lte.cha

A template channel table is present so the file structure validates, but with `out_tot = 0` on the HRU, no flow reaches a channel and the channel outputs will be zero.

Reference: *.cha.

Run the model

```
cd path/to/Osu_1hru
./swatplus-<version>-linux-x86_64
```

The run takes a few seconds because there is only one HRU. Watch the console for `Original Simulation` lines.

Read the outputs

simulation.out

The last line should be `Execution successfully completed`.

diagnostics.out

Every line is a missing reference. Expect a long list because the Editor wrote many template channel and aquifer rows whose companion property records do not all exist. The HRU run itself is unaffected.

Reference: Checker and diagnostics.

checker.out

A single row for `PadHOEGOG`. Confirm `zmx`, `sumfc`, `sumul`, `hydgrp` are reasonable for a paddy.

basin_wb_aa.txt and basin_wb_yr.txt

Annual and average-annual water balance for the basin. Because there is one HRU, basin equals HRU. Compare `basin_wb_yr.txt` (per-year values) to `basin_wb_aa.txt` (the average over years 4 through 14, since `nyskip = 3`).

hru_wb_aa.txt

The water-balance row for HRU 1. Look at `irr` (irrigation depth, large for a paddy), `surq_gen` (surface runoff), `et`, `perc`.

Plant outputs

`crop_yld_aa.txt` reports rice yields by HRU and crop. `hru_pw_aa.txt` carries plant biomass and water-stress days.

What to change next

- Switch to Hargreaves PET (`pet = 2` in `codes.bsn`). The model will stop reading the solar, humidity, and wind files.
- Turn on daily output for `basin_wb` to inspect the paddy fill and drain pattern through the season.
- Change the simulation period in `time.sim` to a single year for a fast iteration loop.

See the cookbook for these edits.

The SWAT+ source tree ships one analysis notebook: `jupyter_notebooks/carbon.ipynb`. It reads the carbon-output files for a single HRU and draws time-series plots of every carbon pool the C-FARM / Century model tracks. Use it after running a project with the carbon model enabled.

What it reads

The notebook reads the following files from a SWAT+ output folder:

- `hru-data.hru` and `landuse.lum`. Used to look up the soil name, the `lu_mgt` name, and the management identifier for a chosen HRU.
- `mgt_out.txt`. Operation log: planting, harvest, tillage, fertilization. The notebook overlays these as event markers on every plot.
- `hru_cbn_lyr.txt`. Total soil carbon per soil layer.
- `hru_seq_lyr.txt`. Sequestered carbon per soil layer.
- `hru_rsdc_stat.txt`. Surface residue and soil carbon pools.
- `hru_plc_stat.txt`. Plant carbon pools.
- `hru_cpool_stat.txt`. Residue and soil carbon pools by layer (metabolic, structural, lignin, microbial, slow humus, passive humus).
- `hru_org_trans_vars.txt`. Organic-matter transformation variables.
- `hru_cflux_stat.txt`. Carbon flux statistics including CO2.

The notebook expects these to be produced by a run where `print.prt` lists each of those output rows with at least one frequency turned on. Daily output is what the notebook is designed for.

What it draws

The notebook is organised by output file. Each section opens with an "Initialize" cell that loads the file into a DataFrame, then one or more graph cells produce a Plotly figure. Sections, in order:

1. Total carbon by layer depth (`hru_cbn_lyr`).
2. Total sequestered carbon and 300 mm sequestered (`hru_seq_lyr`).
3. Surface residue and soil carbon pools at 10 mm depth (`hru_rsdc_stat`).
4. Soil carbon pools at a user-specified depth.
5. Live root mass by soil layer.
6. Plant stats from `hru_plc_stat`.
7. Residue and soil carbon pools (`hru_cpool_stat`) at 10 mm.
8. Same pools at a user-specified depth.
9. Soil residue, metabolic, structural, lignin, microbial, slow humus, and passive humus pools by soil layer.
10. Soil water content by soil layer.
11. Organic-matter transformation variables (`hru_org_trans_vars`).
12. Carbon fluxes (`hru_cflux_stat`), non-CO2 and CO2.

Every plot shows the chosen variable as a time series. PLANT, HARVEST, and TILLAGE events from `mgt_out.txt` are overlaid as vertical markers so you can read the management context off the figure.

How to run it

1. Run SWAT+ on a project with the carbon model on. In `codes.bsn`, set `carbon = 2` to activate the dynamic CENTURY/SWAT-C model (`carbon = 1` is the reserved C-FARM slot and runs no carbon model). The `Ames_sub1` reference dataset ships `carbon = 1`, so change it to `carbon = 2`.
2. Turn on the relevant outputs in `print.prt`. At minimum, `hru_cb` and the carbon-pool outputs need `daily` set to `y`.
3. Open `carbon.ipynb` in JupyterLab or VS Code.
4. Edit the first code cell. Set:
   - `hru_id` to the HRU you want to plot.
   - `wdir` and `sdir` so that `f"{wdir}/{sdir}"` is the folder that holds the SWAT+ outputs.
5. Run all cells (Run All).

The notebook uses `pandas`, `plotly`, and the standard library. Install with:

```
pip install pandas plotly
```

What you should see

Each graph cell renders a Plotly figure inline. Event markers from `mgt_out.txt` line up the plots with the management schedule. The "Initialize" cells print "File not found" and stop rendering that section if the corresponding output file is missing; this is expected when an output is not enabled in `print.prt`.

Related

- Theory: Carbon covers what the pools and fluxes represent.
- print.prt for the output switches.
- Cookbook: enabling the carbon model for the input edits.

Five common edits to a working SWAT+ project. Each recipe lists the file to open, the line to change, and what to check after running. Every recipe assumes you start from a project that already runs successfully.

Add a new HRU

You need a new entry in `hru.con`, `hru-data.hru`, and a row in any companion table the new HRU points to that does not yet contain the records (`topography.hyd`, `hydrology.hyd`, `soils.sol`, …).

1. Open `hru.con`. Append a new row, copying an existing row and bumping the `id`. Edit `area`, `lat`, `lon`, `elev`, and `hru` (the HRU index). Keep `wst` pointed at an existing weather station.
2. Open `hru-data.hru`. Append a row with the same `id` and a new `name`. Point its `topo`, `hydro`, `soil`, `lu_mgt`, `soil_plant_init`, `surf_stor`, `snow`, `field` columns at names that exist in the corresponding tables.
3. Open `object.cnt` and update the `hru` count and `obj` count to reflect the new total.
4. Run. After the run, check `checker.out` for a new row matching the soil name. If `usle_k`, `zmx`, `sumfc` look wrong, your `soils.sol` row is incomplete.

References: hru-data.hru, *.con, object.cnt.

Swap the PET method

`codes.bsn`, field 3 (`pet`):

- `0`: Priestley-Taylor.
- `1`: Penman-Monteith.
- `2`: Hargreaves.
- `3`: read from `pet.cli`.

Edit the value, save, run.

Penman-Monteith needs solar radiation, relative humidity, and wind. If those climate files are absent, the weather generator will fill them in, and the result will not match a station-driven Penman-Monteith run.

Hargreaves needs only minimum and maximum temperature. It is the right choice for sites with minimal climate data.

If you set `pet = 3`, also point `pet_file` (field 1 of `codes.bsn`) at a valid `pet.cli` and provide the daily-PET file it references.

Reference: codes.bsn.

Switch to Green-Ampt infiltration

`codes.bsn`, field 21 (`gampt`):

- `0`: curve-number infiltration (default).
- `1`: Green-Ampt with Mein-Larson.

Green-Ampt runs on a sub-daily time step. Set `step` in `time.sim` to a non-zero value (the source supports `step = 24` for hourly). Your precipitation file must contain sub-daily values; a daily `.pcp` will not be disaggregated for you.

After the run, look at `basin_wb_aa.txt`. The split between `surq_gen` and `perc` shifts when the method changes; if it does not, the run is still using curve number.

References: codes.bsn, time.sim.

Change the simulation period

`time.sim`:

```
day_start  yrc_start  day_end  yrc_end  step
       0       1975        0     2020       0
```

- `day_start = 0`: start January 1 of `yrc_start`.
- `day_start = N > 0`: start on Julian day N of `yrc_start`.
- `day_end = 0`: end December 31 of `yrc_end`.
- `day_end = N > 0`: end on Julian day N of `yrc_end`.

To shorten a run, change `yrc_start` and `yrc_end`. To set a warm-up that is excluded from the average-annual outputs, leave `time.sim` alone and set `nyskip` in `print.prt`.

Your climate files must contain data for every day of the simulation period. If they do not, SWAT+ writes a "file not found" or runs off the end of the file.

References: time.sim, print.prt.

Enable the carbon model

`codes.bsn`, field 14 (`carbon`):

- `0`: static (legacy mineralization).
- `1`: C-FARM.
- `2`: Century.

For `1` or `2`, also:

1. Provide a `carb_coefs.cbn` file in the project folder. The Ames_sub1 dataset ships one you can copy. If `cswat /= 0` and no `carb_coefs.cbn` is present, SWAT+ uses internal defaults and writes a notice to `diagnostics.out`.
2. Confirm `soil_plant.ini` (or `soils.sol`, depending on tool version) carries initial organic carbon by layer. Without realistic initial pools, the first decade of output is transient as pools spin up.
3. Turn on carbon outputs in `print.prt`. Set `hru_cb` and the per-pool outputs (`hru_cbn_lyr`, `hru_cpool_stat`, `hru_cflux_stat`, …) to `daily` if you want to use the carbon notebook.

After the run, `basin_carbon_aa.txt`, `hru_carbon_aa.txt`, and the per-pool files appear in the output folder. If they are missing, the print switches were not on.

References: codes.bsn, carb_coefs.cbn, print.prt.